

Home Credit Default Risk

Darren Black, Chris McTeague, Zac Mendenhall

Introduction

- Who is Home Credit?

Introduction

- Who is Home Credit?



- Financial Risk Assessors

Introduction

- Who is Home Credit? 
- Financial Risk Assessors
- What do they do?

Introduction

- Who is Home Credit?  • Financial Risk Assessors
- What do they do?  • Client Risk Evaluation

Introduction

- Who is Home Credit?  • Financial Risk Assessors
- What do they do?  • Client Risk Evaluation
- What is their goal?

Introduction

- Who is Home Credit?  • Financial Risk Assessors
- What do they do?  • Client Risk Evaluation
- What is their goal?  • **MAKE MONEY!!!**



Empowering
people to live
the life they
want now



Serving the unbanked

Empowering people to enter
the broader global economy



Financial inclusion

Focusing on delivering
greater access to financial
services through financial
inclusion efforts across our
markets

Source: <https://www.homecredit.net/>

Business Problems

- Individuals with limited credit history struggle finding loans

Business Problems

- Individuals with limited credit history struggle finding loans
- Home Credit aims to increase financial inclusion

Business Problems

- Individuals with limited credit history struggle finding loans
- Home Credit aims to increase financial inclusion
- Predict loan repayment ability to expand reach while managing risks.

Objectives

- Identify high risk clients

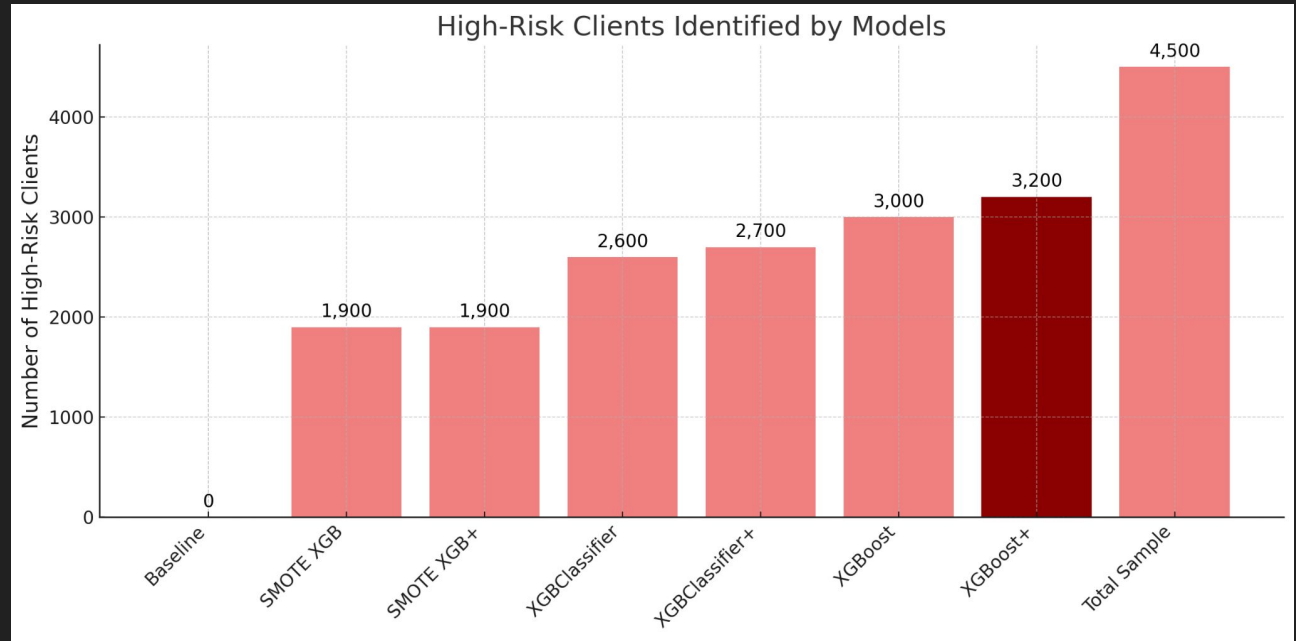
Objectives

- Identify high risk clients
- Identify clients with adequate repayment abilities

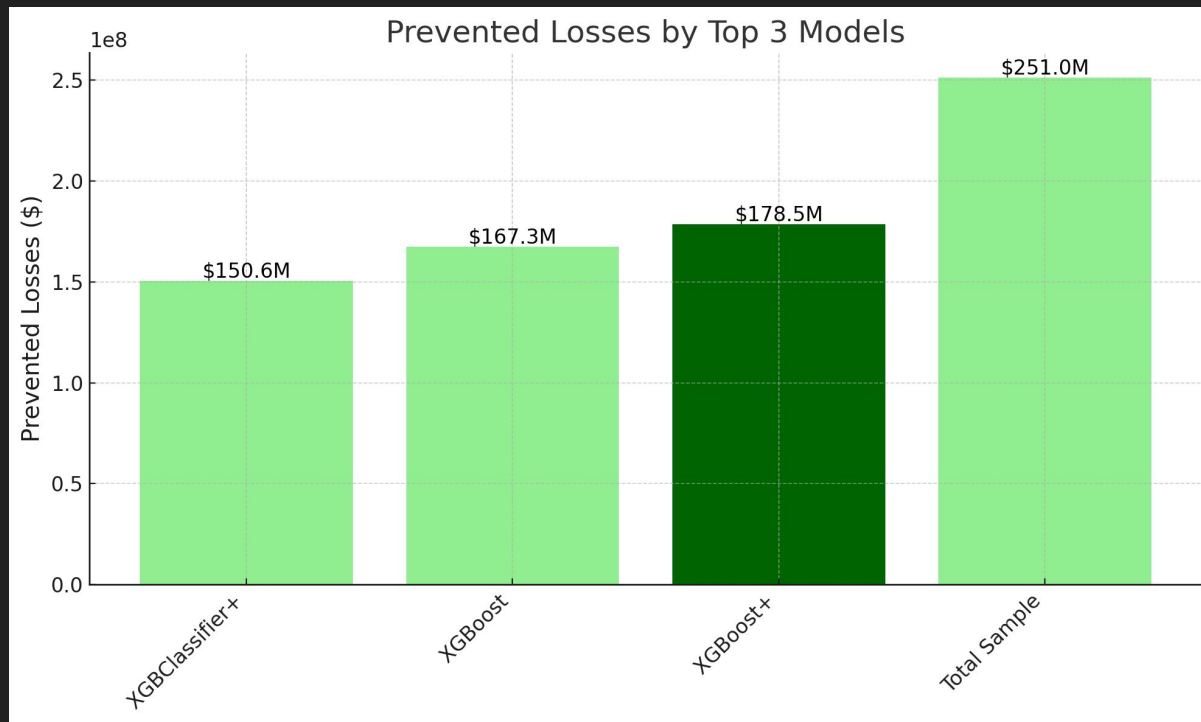
Objectives

- Identify high risk clients
- Identify clients with adequate repayment abilities
- Increase reliability of Home Credit's decision process

- Maximize our ability to identify High Risk clients.
- People can always re-apply
- Recall rate of .75



- Average Loan Amount 557,778
- Reduces Potential Losses to 72.5M, 3.62B Extrapolated Out
- Increased Trust with Lenders



Recommendations

- Apply XGBoost and XGBoost+ models to all potential borrowers

Recommendations

- Apply XGBoost and XGBoost+ models to all potential borrowers
- High Risk Borrowers → stricter approval criteria (higher collateral requirements, lower loan amounts, higher interest rates)

Recommendations

- Apply XGBoost and XGBoost+ models to potential borrowers
- High Risk Borrowers → stricter approval criteria (higher collateral requirements, lower loan amounts, higher interest rates)
- Medium Risk Borrowers → tailored options (shorter loan terms, additional financial checks before approval)

Recommendations

- Apply XGBoost and XGBoost+ models to potential borrowers
- High Risk Borrowers → stricter approval criteria (higher collateral requirements, lower loan amounts, higher interest rates)
- Medium Risk Borrowers → tailored options (shorter loan terms, additional financial checks before approval)
- Improved/Required Customer Education (financial literacy programs)